Curriculum Graph

Mini Project for DS3294: Data Science Practice

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1 Introduction

We developed Curriculum Graph, a system designed to model course dependencies and identify valid academic pathways for students aiming to fulfill major or minor requirements while adhering to other university constraints. The current implementation is tailored to the IISER Pune curriculum.

2 Related Work

Traditional academic handbooks often present course requirements in text-based formats that can be difficult to navigate. This project uses directed acyclic graphs to visualize prerequisites and valid pathways, similar to modern graph software.

3 Our Contribution

We has developed a framework for academic planning that includes:

- A standardized JSON Schema for academic data, including courses, major or minor requirements, and college-wide constraints.
- An interactive Streamlit-based UI [2] utilizing the yFiles widget [3] for graph visualization, manipulation and path filtering.

It can help students answer the following questions:

- What should I take next semester given my completed courses?
- Which prerequisites are blocking my progress?
- Is my current pathway valid for my major and minor?
- Can I still reach the degree target under term constraints?

4 Methods

4.1 Data

The system uses three primary JSON inputs:

- Information on all courses: course code, name, credits, subject, kind, prerequisites, and semesters offered.

Table 1: Dataset and system summary statistics.

Metric	Value
Total courses	264
Program entries (major/minor)	16
Subjects covered	9
Prerequisite edges	94
Average credits per course	3.42
Missing prerequisite links	0
Unknown requirement-code hits	0

- Major and minor criteria: set-wise constraints and exclusions.
- College constraints: number of semesters, minimum credits for degree, compulsory courses each semester and min and max credits per semester.

For IISER-P specifically, we have extracted data from the Curriculum handbook [1].

4.2 Graph Modeling and Planning

We build a directed graph $G = (V, E)$ where V is the selected course set and E contains edges (u, v) whenever course u is a prerequisite of course v. The planning engine uses semester offerings and prerequisite satisfaction to select feasible courses term-by-term.

At a high level, the semester planner greedily selects available courses while respecting per-term caps and prioritizing compulsory requirements.

5 Results

5.1 Quantitative Summary

Table 1 reports key metrics from the latest run.

5.2 Visual Analytics and Plots

Figure 2 shows the course distribution by semester and year, while Fig. 3 shows subject-level and prerequisite structure patterns.

5.3 Dashboard

The primary output is a Streamlit server featuring an interactive graph display. The UI allows users to:

- Visualize all courses with prerequisites, grouped by semester.
- Set interactive options for major, minor, and personal preferences.
- View all valid pathways based on the options chosen by the user.

5.4 USe Case

A student can use this tool to set a major and minor and other preferences. They can view exactly which prerequisites are required and how they fit into their existing corses. The system ensures that credit limits per semester are not exceeded.

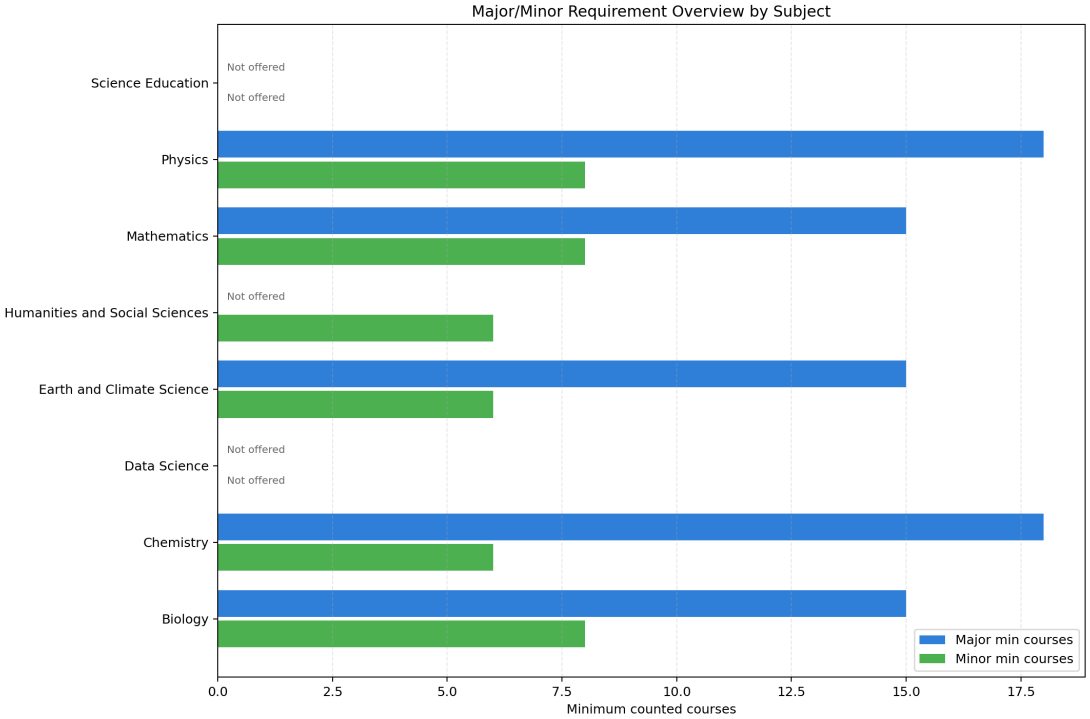

Figure 1: Overview of major-minor requirement structure used in the planning system.

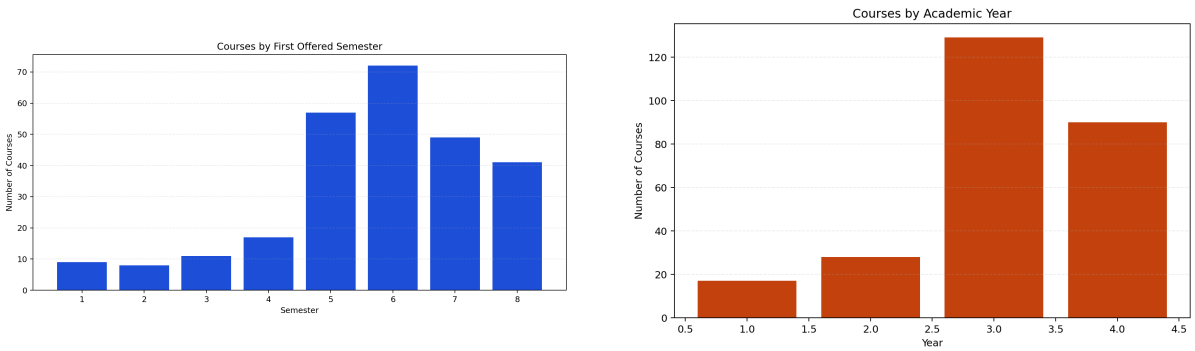

Figure 2: Course availability distribution over semester and academic year.

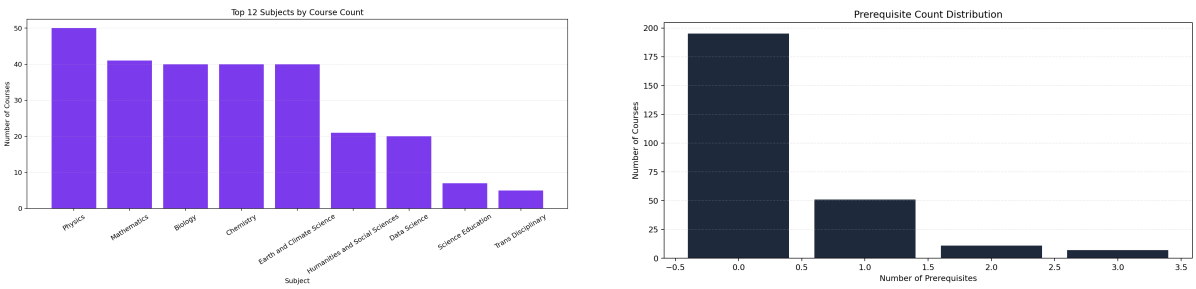

Figure 3: Subject concentration and prerequisite count structure.

6 Conclusion and Outlook

The Curriculum Graph system provides an interactive solution for academic planning. By representing dependencies as a directed acyclic graph, it offers students a clear, customizable roadmap to finish their degree requirements.

7 Acknowledgements

We acknowledge the course and program-rule data sources provided for IISER Pune curriculum modeling, and the open-source tools that enabled this work.

8 Author Contributions

The project responsibilities were distributed as follows:

- **Ayush Raj**: Data Validation, Graph Construction, and UI Testing.
- **Applen Tahu**: Data Extraction, JSON Schema design, and UI Design.
- **Malhar Patel**: Graph Algorithms and UI Construction.
- **Sankar Maharajan**: Data Preprocessing and Graph Validation.

9 Code availability

Source code (GitHub): https://github.com/Mal-Pat/curriculum-graph.

10 References

References

[1] IISER Pune. Iiser pune curriculum data files (course catalog, major/minor requirements, and constraints), 2026. Project dataset used in this report.

[2] Streamlit Inc. Streamlit documentation. https://docs.streamlit.io/, 2026. Accessed: 2026-04-03.

[3] yWorks GmbH. yfiles graphs for streamlit. https://github.com/yWorks/yfiles-for-html-demos/tree/master/demos/toolkit/streamlit, 2026. Accessed: 2026-04-03.